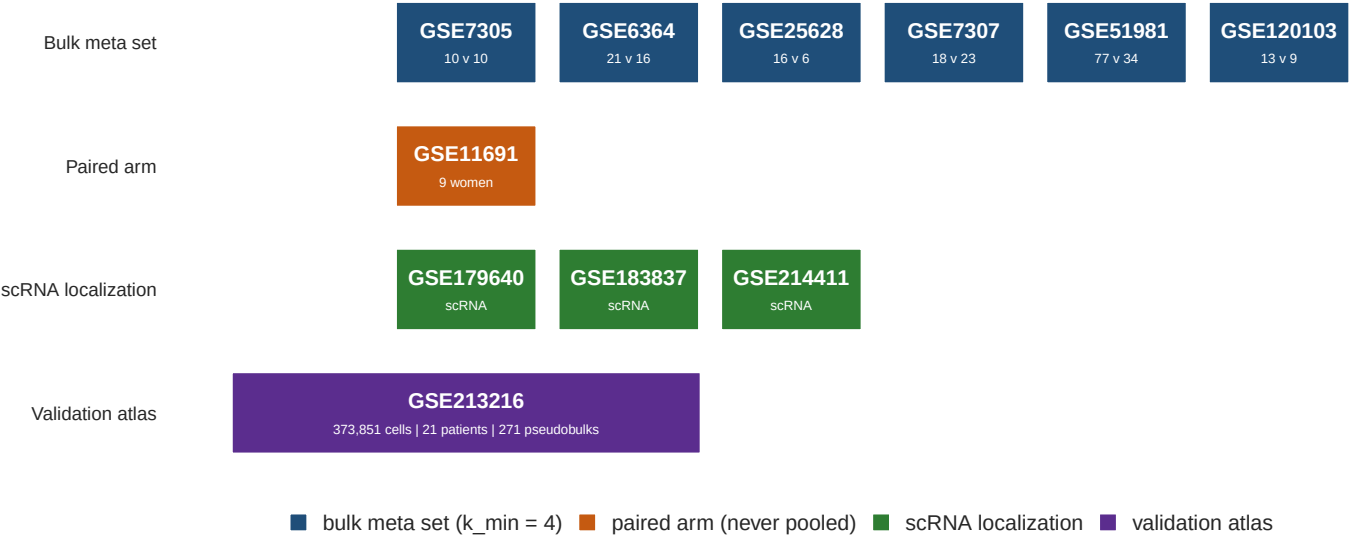
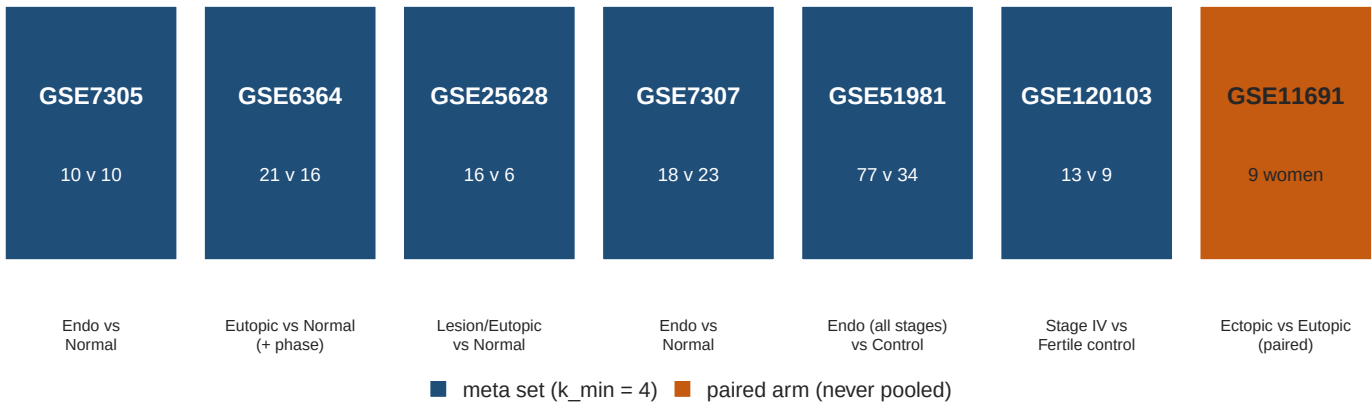


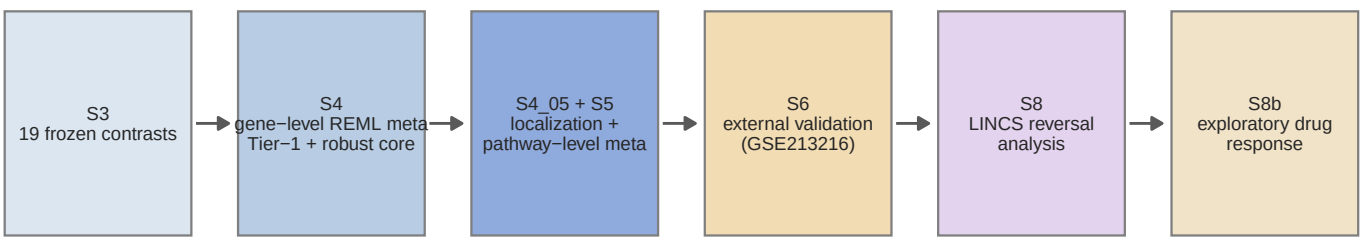
Dataset map
six bulk cohorts (meta set); one paired arm; three scRNA localization datasets; one independent validation atlas



Locked contrasts: six cohorts (meta set) + one paired arm
six disease-vs-control cohorts enter the random-effects meta; the paired ectopic-vs-eutopic arm is reported separately and never pooled



Locked analysis pipeline



Analytical branches
discovery to validation to reversal; BH-FDR < 0.05 is the sole claim gate; exploratory analyses carry no claim

